Global CO_2 Emissions: Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024\nData Sources: Global Carbon Budget 2025, NASA

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

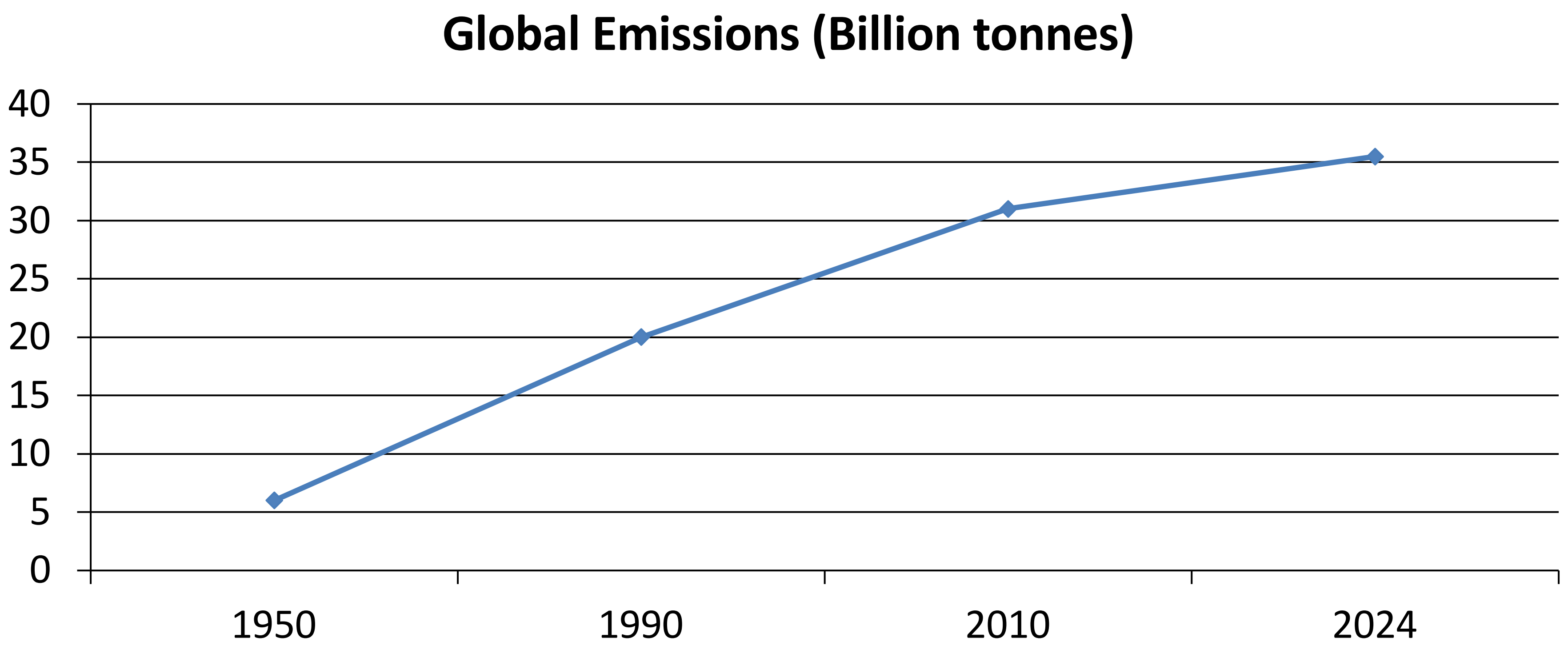
- Latest Measurement (Feb 2026, Mauna Loa): 429 ppm
- Pre-industrial baseline: ~280 ppm (more than 50% increase)
- The current level is at least 1.5 times greater than the natural increase observed at the end of the last ice age 20,000 years ago.
  - This massive shift took roughly two centuries instead of thousands of years.
- Linked primarily to the burning of coal, oil,

429 ppm

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

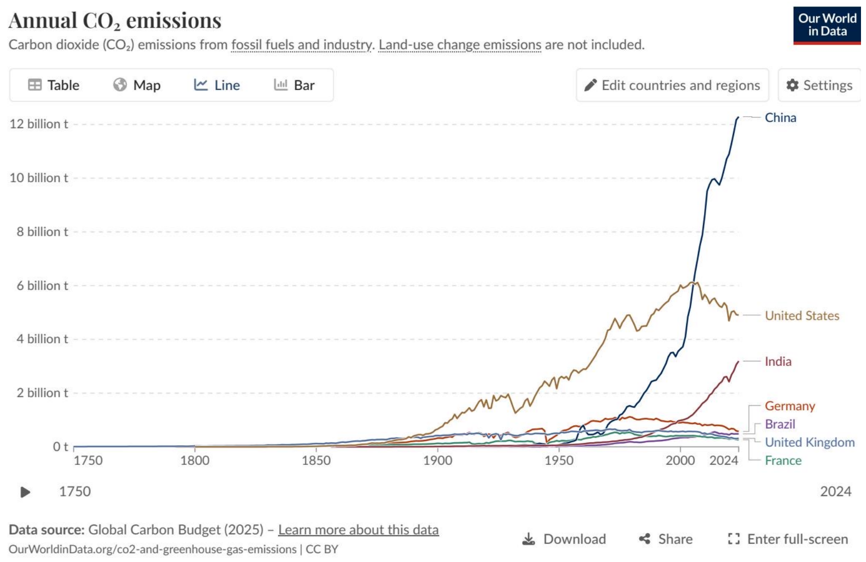
Historical Trajectory (1750–2024)

Before the Industrial Revolution, CO_2 emissions were negligible. Growth remained relatively slow until the mid-20th century.



China Now Emits Over 12 Billion Tonnes — More Than Double the

Country / Region	Annual CO_2 (Bt)	Global Share	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
Brazil	~0.5	~1.4%	Roughly stable
United Kingdom	~0.3	~0.9%	Declining
France	~0.3	~0.8%	Declining



Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

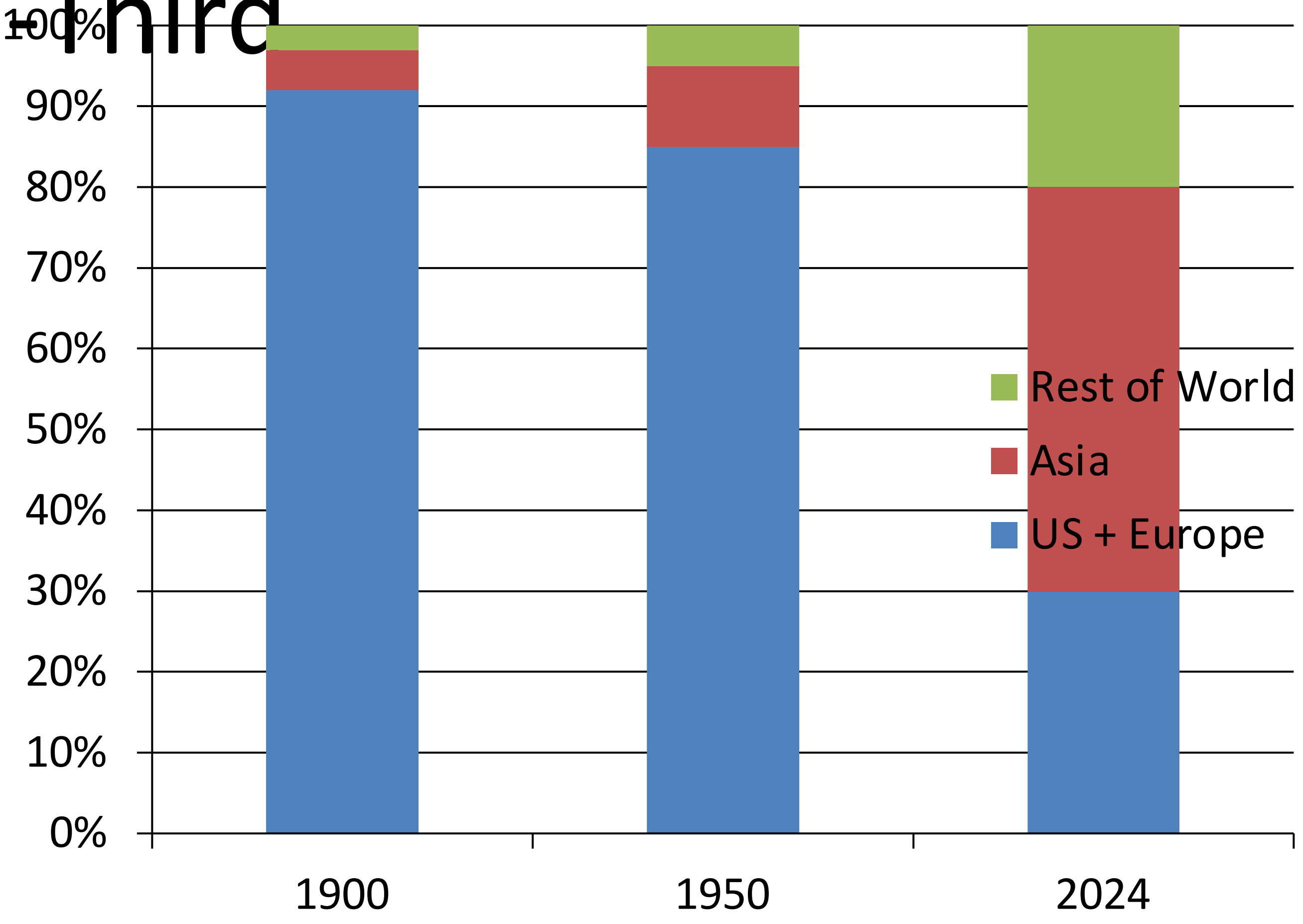
Dramatic Regional Shift:

• 1900: Europe and US accounted for >90% of global emissions.

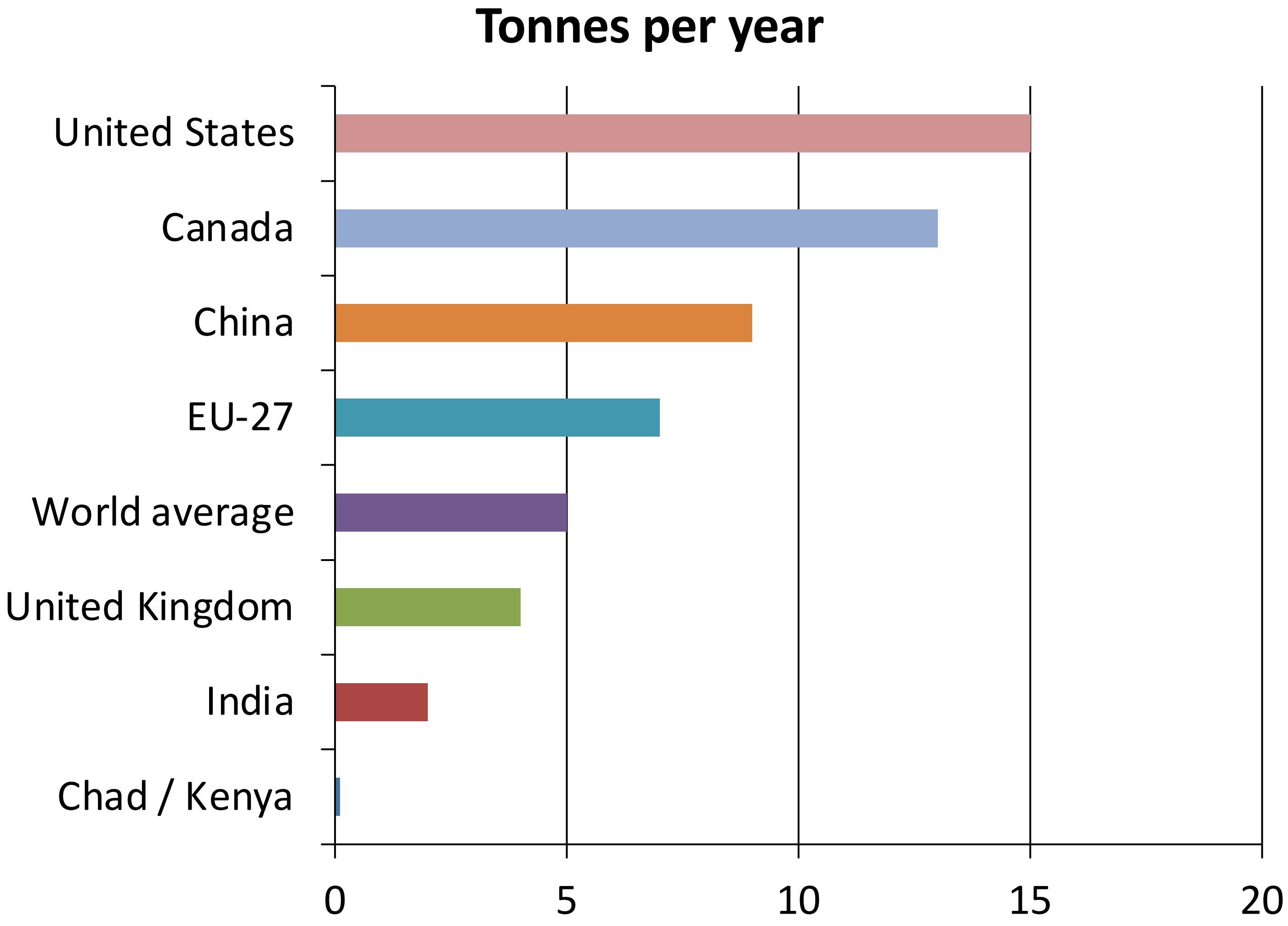
• 1950: Europe and US still >85%.

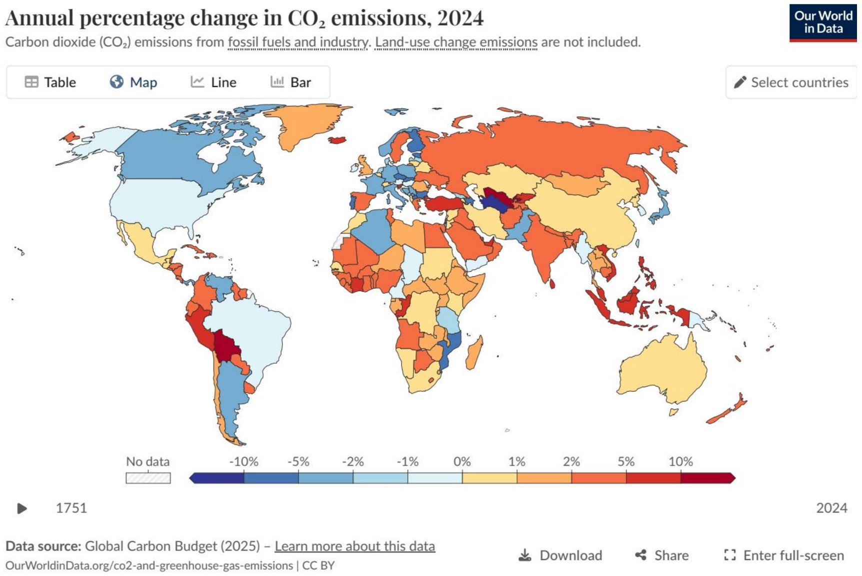
• Today: Asia produces ~50% of all emissions, while US + Europe combined are <33%.

• Africa and South America: each contribute only 3-4%.



Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity





2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

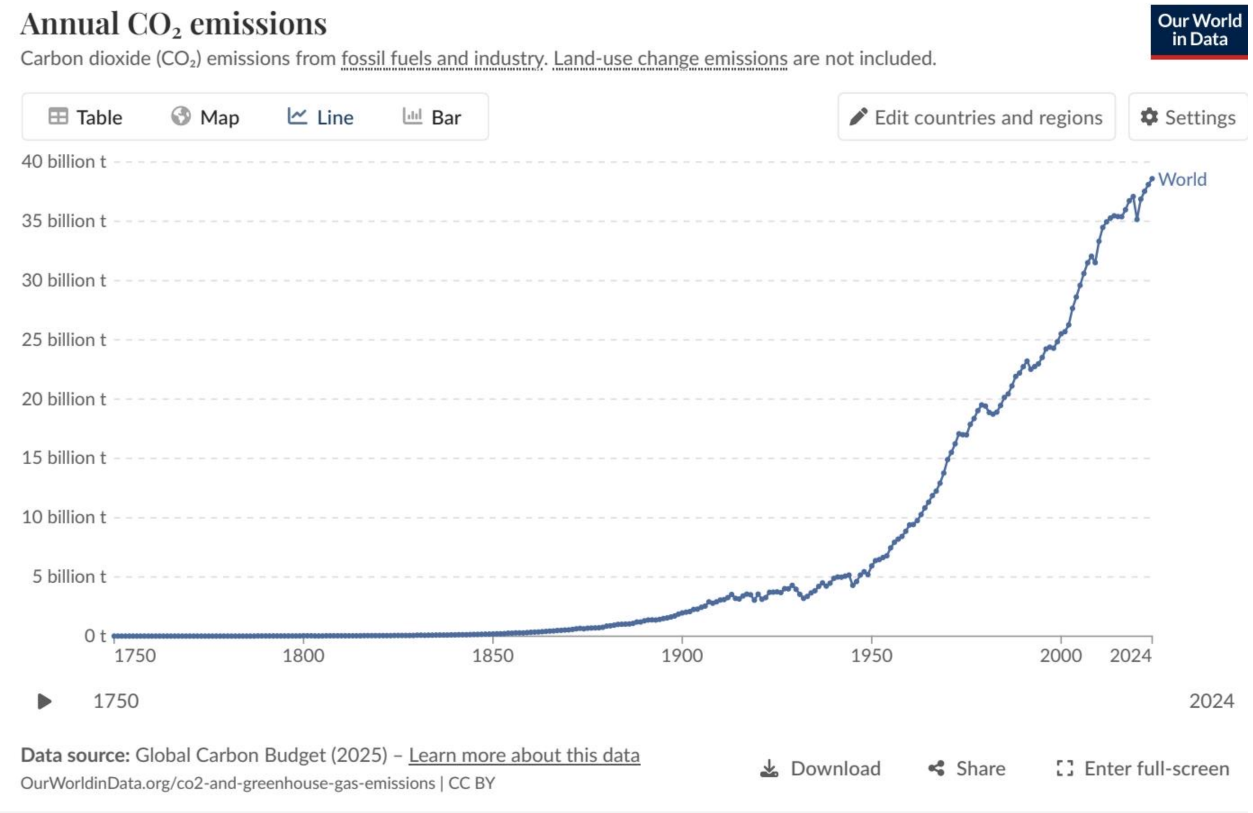
2024 Annual Change Highlights:

• US & Europe: Recorded declines (-1% to -5%), shown in blue on the global map.

• Russia & SE Asia: Saw noticeable increases (+2% to +10% or more), shown in orange/red.

• Africa: Some increases (+2% to +10%).

• South America: A mixed picture with some very large increases (>10%) alongside declines.



Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar prosperity levels can differ significantly in per capita emissions due to their reliance on nuclear, renewables, or fossil fuels.

Country	Per Capita CO_2 (tonnes/year)	Primary Energy Driver / Mix
France	~4	High share of nuclear and renewable energy
United Kingdom	~4	Phase-out of coal, increased offshore wind/renewables
Germany	~8	Higher historical reliance on coal/fossil fuels, industrial base
Netherlands	~8	High reliance on natural gas and fossil fuels

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- Global emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
- China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is rising fast.
- Per capita inequalities remain extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).
- US and Europe emissions are declining, but their historical cumulative responsibility remains large as the geographic center shifts to Asia.
- Policy and energy mix choices — not just prosperity — determine per capita outcomes, as seen in differences between France and Germany.